

Affective transfer between LLM agents: the estimator, two pre-registered ablations, and a control that bites

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Abstract

When two language-model agents converse and one is steered into an emotion, the other’s internal representation of its own state moves with the dose: five of six emotions on Qwen3.6-27B, in one sampling replication and two bit-identical pipeline reproductions on one pool, and the same five on Llama-3-8B (**b1f.***, **b1c.***, **b1c8.***). We ask what carries it, and first fix the instrument. The direction most work reads and writes such states with, a logistic-regression coefficient row, does not reproduce across independent fits in the regime everyone works in, and we show why: on these activations every logistic fit is a perfect separator, so the direction is the penalty’s tie-break among separating hyperplanes, drifting away from the mean-difference direction as the sample grows while the mean-difference direction converges (**a2.***, **a2f.***, **a2m.***). Using the reproducible direction, two pre-registered ablations test the linear account of the channel. Removing the emotion direction from the sender’s span at every readable layer leaves a median 91% of the transfer (**b1c.***); removing a rank-5 affect subspace fails its own instrument check, and a label-free subspace with the affect subspace’s footprint, ten times the emotion direction’s, blocks as much as the emotion direction (**b1d.***). A pre-registered fitted, label-free rank-1 control then returns H3: the control is inert and the emotion direction blocks afraid by 25% and sad by 29% against it; a top-principal-direction control with six times the footprint blocks other emotions but not these while removing most of their readable affect, so removal and blocking dissociate for these two emotions (**b1e.***). Only sad’s block replicates across the two models. The lesson for controls: a random direction was inert in every run and so was a fitted rank-1 label-free direction; only a fitted rank-5 frame blocked, and whether rank or footprint does that is untested.

1 Introduction

Steering vectors and linear probes are the working tools of representation-level interpretability [4, 7–9]. They are also the natural instrument for a question about interacting agents: when agent A is pushed into an emotional state and talks to agent B, does anything of that state arrive in B’s own representation, and if so through what? The question is not exotic. Multi-agent systems built from a single model family are becoming common, and the transfer of affect between them is exactly the kind of thing that would be invisible in transcripts and visible in activations.

Answering it requires a direction one can trust twice: once to read B’s state and once to intervene on A’s. This paper is organised around that requirement. Section 3 shows that the logistic-regression direction does not reproduce across independent fits on either of two models, that holding the per-sample penalty fixed does not repair it, and why: at every sample size we used, every

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fit separates its training half perfectly, so the direction is the penalty’s tie-break among separating hyperplanes rather than a likelihood estimate. Difference-of-means directions, an average rather than an extremum, do reproduce. Section 4 establishes the transfer with that direction. Section 5 reports three pre-registered ablation experiments on the channel and a replication on a second model. Section 6 draws the general lesson about controls, and Section 7 lists what is not claimed.

Three things about how this was done matter for reading it. Each ablation experiment was pre-registered with its decision rule before its driver existed, or before its run for the first; each driver was reviewed and smoke-tested by an independent agent before launch; and each result was critiqued blind by an agent that saw only the code, the pre-registration and the result file. Seven of our own interpretive sentences were withdrawn under that review and are absent here; the registry records each withdrawal.

Relation to prior work. That logistic probe directions can be unstable is not our discovery. Marks and Tegmark [5] identified the deficiency and proposed mass-mean probing; Gao et al. [2] name directional stability as a requirement for probe-then-steer pipelines and propose ridge-tuned logistic regression, reporting robustness of 0.88–0.99 under 20% training-data ablations. Our contribution is the quantity: reproducibility across *disjoint* halves, on the same estimator family, is 0.41–0.57 on the 27B (`stabdiag.logreg_27b_600`, `a2.focus.logreg_split_half`) and 0.29 on the 8B (`a2_8b.*`), and it does not rise with the sample size while difference-of-means rises to 0.94–0.98 on the same splits. Two overlapping 80% subsamples share most of their examples; two disjoint halves share none, and the gap between the two numbers is the finding. Braun et al. [1] corroborate from the other side, noting that steering vectors from few samples vary widely, without fitting twice. Herbster et al. [3] report mean-difference and logistic directions as near-identical (0.98–0.99) for compassion; in a quantity closer to theirs than our split-half battery, though not identical to it, we find 0.43 and 0.33–0.37. Raju [6] measure stability of a representation across feature dimensions, a different axis.

2 Setup

Two agents, A and B, converse in 29 fixed scenarios. A’s message is generated under a steering vector added at layer 42 of Qwen3.6-27B (revision `6a9e13bd`) at doses 0, 0.33, 0.67 and 1.0 of a fixed scale, with three replicate seeds; B’s reply is then sampled with common random numbers across experimental arms, so that sampling noise shared between two arms cancels inside each paired contrast. Two readouts: B’s *present-emotion projection*, the projection of B’s reply activations on a probe fitted on one half of a 1,615-item probe pool while steering and ablation directions come from the disjoint other half; and a generation-side forced-choice readout of A’s span, used as the manipulation check. Every dose-response slope is an ordinary least-squares slope over the four doses on per-scenario means; every confidence interval is a scenario-blocked bootstrap (29 clusters, 5,000 draws), and every between-arm contrast is scenario-blocked and dose-paired. The second model is Llama-3-8B-Instruct-abliterated (revision `dd67dd05`), used for the estimator results and for replications of two ablation designs. All code runs from one vendored core; every number below was produced by a committed script from a committed result file.

3 The estimator: reproducibility and its mechanism

Disjoint-half reproducibility. We fit the same estimator on two disjoint halves of n items each, ten independent splits, and report the mean per-class cosine between the two fitted directions

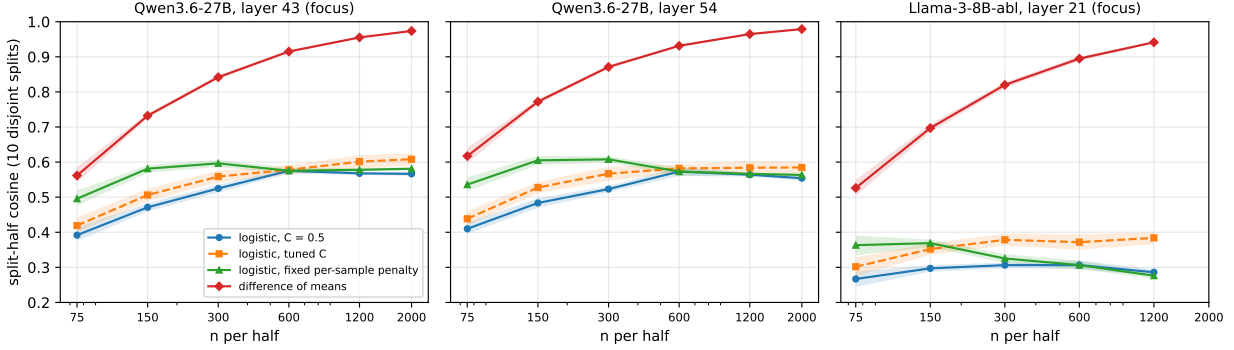


Figure 1: Split-half cosine of the fitted direction over ten disjoint splits as a function of n per half, at two depths of the 27B and the focus layer of the 8B. Difference-of-means rises toward 1; the three logistic variants plateau or fall. Bands are bootstrap CIs over splits.

(Figure 1). At the 27B focus layer with $n = 2000$ per half, difference-of-means reaches 0.974 and logistic regression ($C = 0.5$) 0.566 (`a2.focus_dom_split_half`, `a2.focus_logreg_split_half`); at layer 54, 0.979 against 0.554 (`a2f54.*`); on the 8B focus layer at $n = 1200$, 0.941 against 0.286 (`a2_8b.*`). Tuning C per fit by cross-validation reaches 0.61 on the 27B (`a2f.focus_tuned_c_does_not_close_gap`) and 0.37–0.41 on the 8B.

Holding the per-sample penalty fixed does not help. A natural objection is that a fixed C changes the effective penalty as n grows. With $C_n = 0.5 N_{\text{ref}}/n$, so that the per-sample penalty is constant, the logistic curve shows no net rise from $n = 150$ to 2000 at two anchors on the 27B focus layer (0.581 at both ends for the 600 anchor, with a significant dip at 600 and a partial recovery; `a2f.focus_logreg_fixed_lambda_flat`, `a2f2.lam_anchor2000_no_rise`), one drop at 300→600 and then flat at layer 54 (`a2f54.lam_declines_anchor600`), a rise at layer 16, and on the 8B a peak at 150 followed by a fall to 0.276 at 1200 (`a2f8.lam_declines`). The picture is depth-specific, with no convergence over the reachable range ($n \leq 2000$, $d = 5120$): a proportional-regime statement, not an asymptotic one.

Why. Every multinomial logistic fit on every half at every n used, under the fixed schedule and $C = 0.5$, 50 and 500 alike (the two weak arms being one tolerance-limited solution), separates its training half perfectly: accuracy exactly 1.000 in every cell, arm and seed; worst single-fit log-loss 0.0055; at most 80 iterations (`a2m.separable_every_n`, `a2m.8b_same_regime`, `a2m.v2_ten_seeds`). The likelihood is saturated from $n = 75$: the direction is the penalty’s tie-break among separating hyperplanes. Weakening the penalty a hundredfold leaves the direction unchanged on the same half (cosine ≥ 0.96) and no more reproducible (0.52 against 0.58 at $n = 2000$; `a2m.weak_penalty_same_direction`). The separator starts near the mean-difference direction and departs from it as n grows, cosine 0.97→0.63 on the 27B (`a2m.logistic_departs_from_dom`) and 0.91→0.43 on the 8B (`a2m.8b_same_regime`), while difference-of-means becomes reproducible (Figure 2). The wording adopted from blind review: on separable activations the multinomial direction is the penalty’s tie-break among separating hyperplanes, not a likelihood estimate; its split-half reproducibility plateaus at 0.55–0.60 on the 27B and falls to 0.28 on the 8B across a 27-fold increase in n , while difference-of-means, an average rather than an extremum, rises to 0.94–0.98 on the same splits. Not claimed: that the fit sits at the max-margin limit (the weak arms are tolerance-limited), or a support-vector mechanism for the drift.

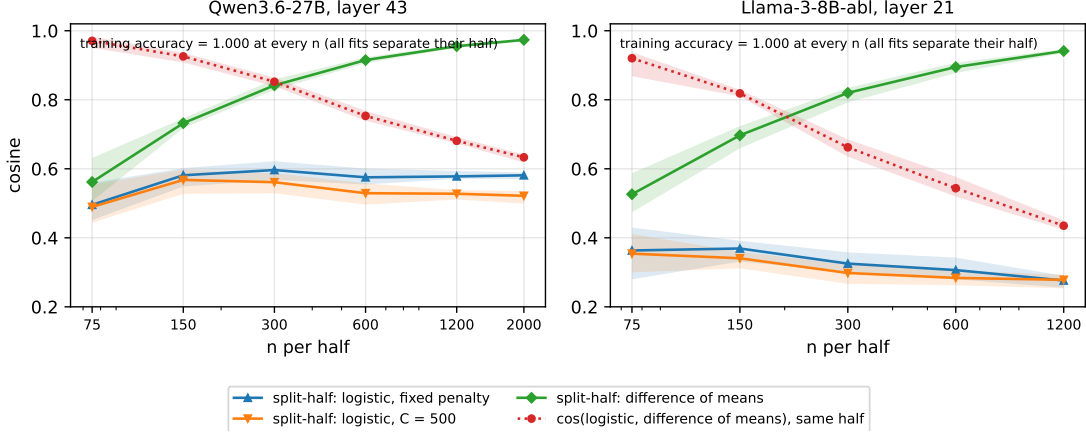


Figure 2: The mechanism. Within one half, the cosine between the fixed-penalty logistic direction and the difference-of-means direction (dotted) falls as n grows, while difference-of-means becomes reproducible across halves (green) and the logistic direction does not, under the fixed schedule or a $1000\times$ weaker penalty alike. Every fit has training accuracy 1.000.

Against the near-identity claim. In a quantity closer to that of Herbster et al. [3] than our split-half battery, though still not identical to it (mean-difference against logistic, raw space, ten sub-samples), the cosine is 0.43 on the 27B and 0.33–0.37 on the 8B, not 0.98 (`a2f.caa_raw_vs_logreg_raw`, `a2f8.cross_estimator`).

4 Transfer

B’s present-emotion projection rises with A’s dose for desperate, afraid, happy, sad and angry, with none-arm slopes from +48 to +212 and CIs excluding zero; calm’s (+44 [−3, +99]) never does (`b1c.verdict_H1`; Figure 3). The same slopes appear, bit for bit, in the three runs that share the pool and directions (`b1c.partial_block_replicates`, `b1d.replicates_b1c_arms`), and the manipulation check on A passes for all six emotions in each. Happy’s is the weakest of the five (+48 [+8, +88]). On the 8B the same five emotions transfer (`b1c8.transfer_replicates`), with a steering manipulation half as strong and less specific, so only the pattern, not the size, is compared across models.

5 Three pre-registered ablations

B1c: the rank-1 direction (verdict H1). Projecting the difference-of-means emotion direction out of A’s span at every hidden state from 13 to 63 during B’s prefill removes a median 9% of the dose-response, CI [4%, 25%] over five testable emotions (`b1c.verdict_H1`, `b1c.median_fraction`). Afraid (21%) and sad (33%) are the only significant blocks (`b1c.partial_block_replicates`). Extending the ablation to layers 43–63 changes no dose-response detectably, with the extra blocking bounded under 0.08 for four emotions, a driver-declared contrast (`b1c.no_gain_beyond_layer42`). A unit-norm random direction, removing twice the norm, leaves five of six slopes within 5% of the none arm and moves happy by +28% with overlapping CIs, descriptively, since no random-versus-none contrast was computed (`b1c.random_control_inert`). The blocked fraction tracks how much readable affect the ablation removes ($r \approx 0.85$ over five points; blocked-per-removed ≈ 0.27 ; `b1c.readout_residual`): the rank-1 direction is insufficient to carry the transfer.

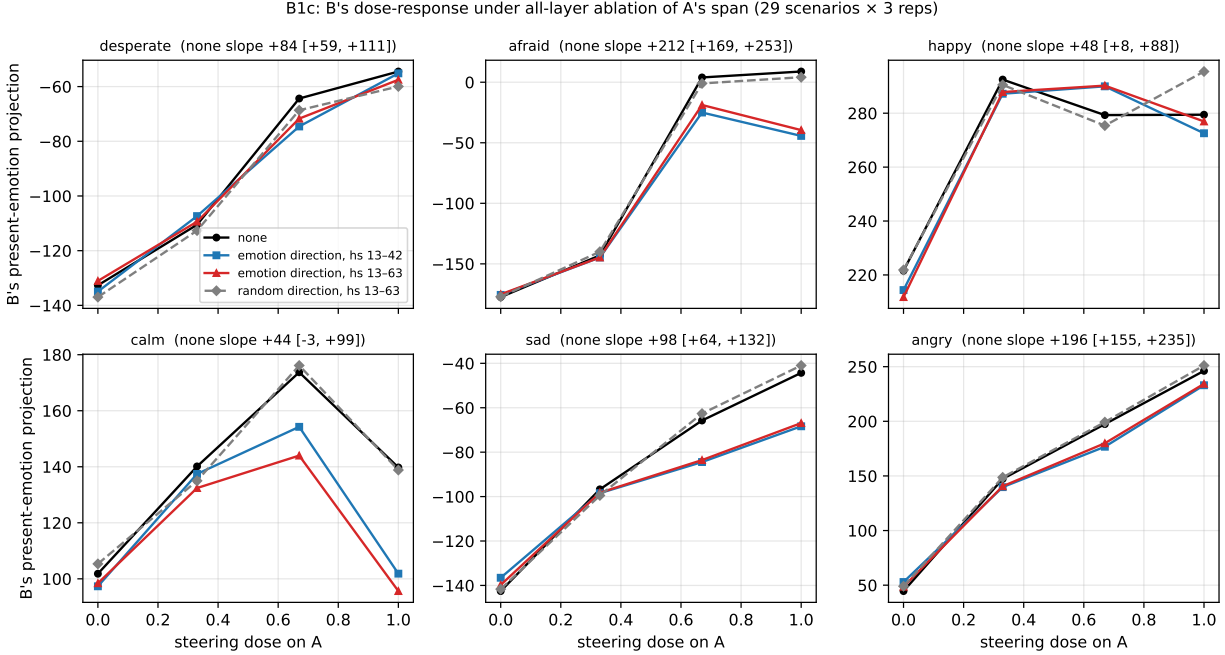


Figure 3: B's present-emotion projection against A's steering dose, per emotion, under no ablation and under the three ablation arms of B1c (29 scenarios $\times$ 3 replicates). Titles give the none-arm slope with its scenario-blocked CI.

B1d: the rank-5 subspace (verdict `instrument_failed`). Projecting out the rank-5 class-mean affect subspace, which captures 16–21% of the pool's variance and has cosine 0.71–0.81 to the top-5 principal components (`b1d.frame_geometry`), removes at least 80% of the readable affect for only three of six emotions and shifts the readout at dose 0 (desperate 0.11 $\rightarrow$ 0.50; `b1d.readout_baseline_shift`): the instrument and the ablation share a target. A permuted-label subspace built by the same estimator, with no emotion information and a third of its span in common with the affect frame, reduces the transfer for afraid (−49.5), sad (−26.7) and angry (−44.3), as much as the emotion direction did (`b1d.perm_control_blocks`; unadjusted CIs), while a random 5-frame is inert. The emotion-specific increment over that control is 0.14 [−0.03, 0.29], and no per-emotion contrast clears the registered BH-FDR rule ($q = 0.051$; `b1d.median_fraction_descriptive`); rank 5 differs from rank 1 in opposite directions, less blocking for afraid (+29.3) and more for angry (−60.6) (`b1d.rank5_vs_rank1`).

B1e: a fitted, label-free rank-1 control (verdict `H3`). The same estimator on permuted labels gives a rank-1 direction with cosine 0.004 to the emotion direction (measured before launch, not stamped in the file) and $0.85\times$ its footprint; removed at all 51 layers it changes no emotion's dose-response and removes no readable affect (`b1e.permdir_inert`). Against it the emotion direction blocks afraid by 25% [17%, 33%] and sad by 29% [7%, 56%] (BH-FDR q 0.001 and 0.028; `b1e.verdict_H3`). The top principal direction of A's span, with $5.8\times$ the footprint, blocks angry by 61% (and calm, knife-edge on an unstable emotion) while leaving afraid and sad untouched even though it removes 79% and 71% of their readable affect (`b1e.pc1_dissociation`): readable-affect removal and blocking dissociate. B1d's rank-5 permuted frame blocks where B1e's rank-1 permuted direction does not; they differ in rank and in footprint (6.5 against 0.5 per position), and which matters is untested. The control's footprint ($0.84\times$ and $0.77\times$ for afraid and sad, below B1c's inert random direction)

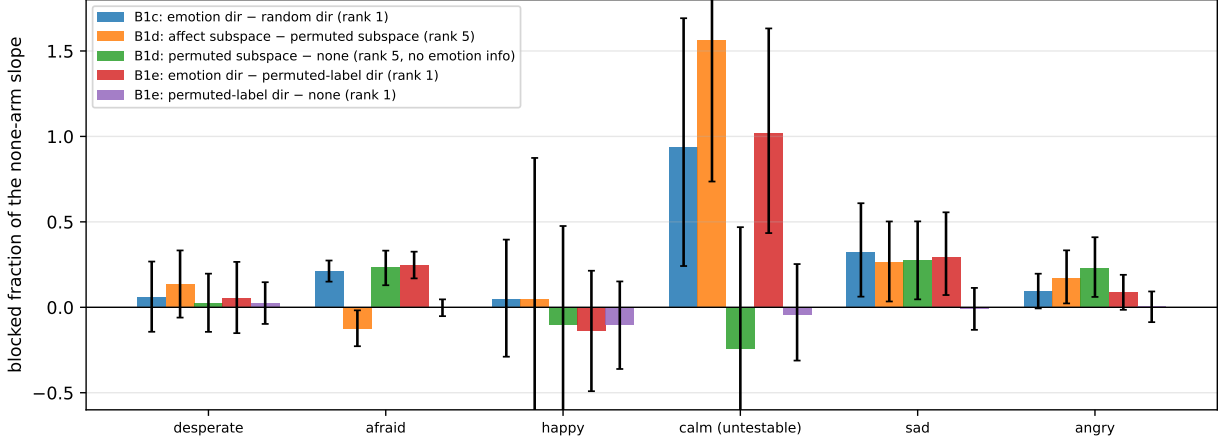


Figure 4: Blocked fraction of the none-arm slope, per emotion, for the decision contrast of each ablation experiment and for the two label-free controls against baseline. Error bars are the CI-implied ranges. Calm is unstable (its none-arm slope includes zero).

Table 1: The three ablation experiments on the 27B, decision contrast per emotion (blocked fraction of the none-arm slope; significant contrasts in bold). Calm is unstable.

	desperate	afraid	happy	calm	sad	angry
B1c: emotion dir – random dir	0.06	0.21	0.05	(0.94)	0.33	0.09
B1d: affect subspace – permuted subspace	0.14	-0.13	0.05	(1.56)	0.26	0.17
B1d: permuted subspace – none	0.03	0.23	-0.10	(-0.24)	0.27	0.23
B1e: emotion dir – permuted-label dir	0.06	0.25	-0.14	(1.02)	0.29	0.09
B1e: permuted-label dir – none	0.02	-0.00	-0.10	(-0.04)	-0.01	0.00

leaves the footprint caveat standing. The secondary readout on B’s reply contradicts afraid’s block (null, wrong sign) and agrees for sad.

Replication on the 8B. B1c’s design blocks the same two emotions to significance on the 8B (15%, 31%; `b1c8.verdict_and_block`), with the other three too wide to tell agreement from disagreement; its manipulation-check rule returned `instrument_failed` because the upper ablation window changes the readout by nothing measurable there and the strict criterion has no tolerance band. B1e’s design returns H3 on the 8B with sad blocked by 36% (q 0.002, secondary readout agreeing) and afraid by 14% (q 0.08, no block against baseline); the control is matched only on average (sad 0.86 $\times$), both directions remove about the isotropic norm, and B1c’s random direction on that model reproduced the contrasts, so the run adds no discriminating power there. Only sad replicates across models on both outcomes; afraid is unreplicated (`b1e8.verdict_H3_matched`).

6 What the controls teach

Norm-matched random and orthogonal directions, the field’s standard controls, were inert in every run; so was a fitted, label-free rank-1 direction. What blocked without emotion information was a fitted rank-5 frame with ten times the emotion direction’s footprint. Any ablation claim about a concept direction therefore needs, beside the random control, a control fitted by the same estimator

on permuted labels, and if the intervention is more than rank 1, a control matched in rank and footprint; and the manipulation check must not share its target with the ablation, or the check fails in the way B1d’s did.

7 Limitations and what is not claimed

One probe pool and one direction estimate underlie all ablation runs on the 27B; the A-span readout is forced-choice and shares a target with the subspace ablation; calm is untestable at this dose grid; B1d’s per-emotion contrasts do not clear BH-FDR; the median CIs are scenario-only bootstraps and do not include emotion-to-emotion spread. Provenance limits: B1c’s pre-registration stamp is empty in its result files and its ordering rests on commit times, file modification times and the run’s start time; B1c’s pool is 1,615 against the registered 2,160 and its FDR ran over five emotions rather than six; the 8B runs of the B1c and B1e designs are out-of-pre-registration replications; B1e’s control cosine was measured by the pre-launch verifier and is not stamped in the file; the weak-penalty arms are tolerance-limited and bound nothing about the $C \rightarrow \infty$ limit. Not tested: the token channel (three rewrite designs failed their own manipulation check: the neutral rewrite kept 67–98% of the readable affect and the affect-preserving rewrite changed the transfer itself; `b1c.text_arms_uninterpretable`), the behavioural reach, layers below 13, and $n > d$.

8 Practice

Pre-registration before each run (B1d and B1e stamped in their files and written before their drivers; B1c’s written after its driver, before its run, and unstamped), an independent pre-launch verifier with a forward-pass smoke test, and a blind critique after every result whose corrections are recorded in the repository’s critique ledger. Seven of the authors’ interpretive sentences were withdrawn under review and are absent here.

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